

**“Build a simple low power Step it up or
down (DC-DC Buck Boost Converter)
using LT-Spice”**

Aim

Basic understanding about LT-Spice

LT-Spice simulation installation instructions

Connections - Circuit Diagram - Detailed Steps

Task - Exercise

Aim

Very often a low power buck boost converter used to convert some low voltage to a higher voltage or higher voltage to lower voltage is required in a system. This build exercise is to learn to build one such simple converter, using LT-Spice simulation in an open loop.

LT-Spice - Introduction

LT-Spice is an open source, high-performance simulation tool developed by Analog Devices. It is used to simulate, design, and test electronic circuits before physically building them.

- **Why LT-Spice is important?**

LT-Spice provides a safe learning environment where peoples can freely experiment with electronic circuits without risk of damaging physical components. It helps them to understand circuit behaviour by allowing real time visualization of voltage, current, and other parameters. By simulating before building, Peoples save time and cost as they as they do not need to buy multiple components or PCBs initially. LT-Spice also encourages peoples to develop strong problem -solving skills, teaching them how to identify and fix circuit issues logically. It offers valuable hands-on simulation experience that connects theoretical concepts learned in classrooms with practical real-world electronics. Most importantly,

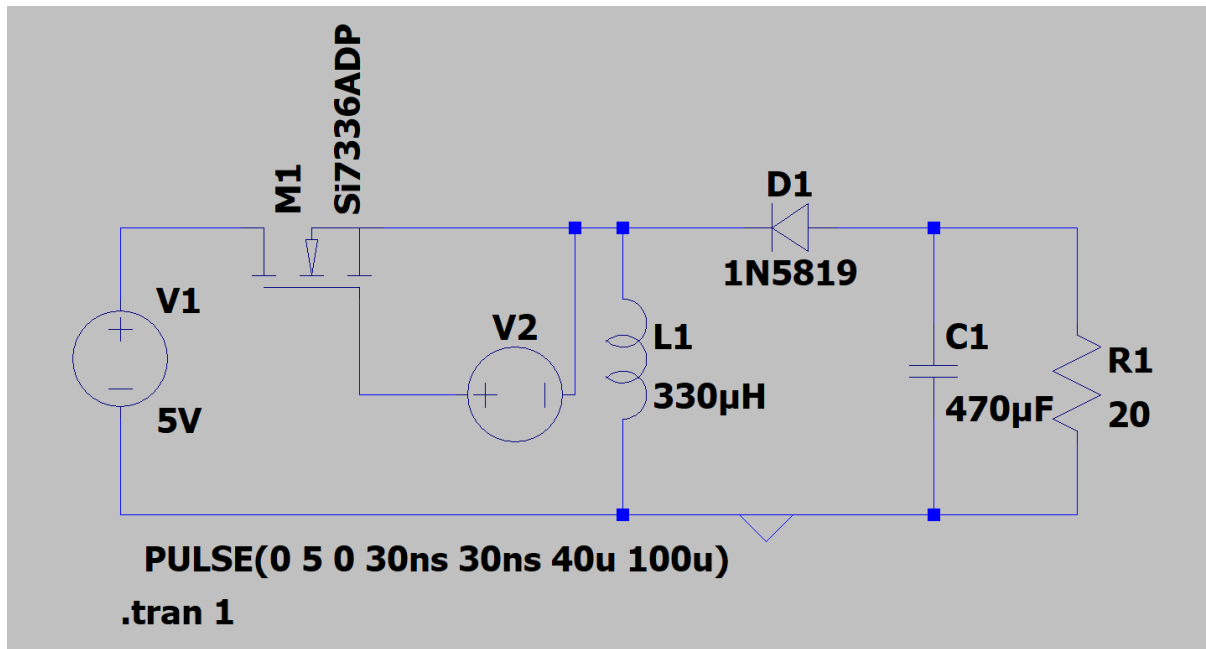
learning LT-Spice gives peoples and industry-relevant skill, preparing them for careers in research, development, and engineering industries.

LT-Spice Simulation Installation Instructions

- 1) **Open your web browser** (Such as Chrome, Firefox, or Edge).
- 2) In the search bar, type **LT-Spice** and press Enter.
- 3) Click on the official link from **Analog devices**, usually titled something like “**LTspice**” | **Analog devices**”. Or use this hyperlink to download [LT-Spice](#).
- 4) **On the LTspice page**, look for a **Download** button or section.
- 5) **Choose the correct version** for your operating system (Windows, macOS, etc.).
- 6) **Download the installer file** to your computer.
- 7) **Once the download is complete**, open the installer file.
- 8) **Follow the on-screen instructions** to install LT-Spice:
 - Accept the License Agreement
 - Choose the installation folder (or use the default)
 - Complete the installation process.
- 9) **After installation, open LT-Spice** from your desktop or Start Menu.
- 10) **Now are you ready to start using LT-Spice for circuit simulation.**

Connections

Circuit Diagram:



Detailed Connection Steps

Step 1: Open LT-Spice

1. Open **LT-Spice** on your laptop.
2. Wait until the main window appears.

Step 2: Create a New Schematic

1. Go to the **File** menu.
2. Click **New Schematic**.
3. A blank schematic window will appear.

Step 3: Place components

We need to place these components one by one:

Component	Symbol in LT-Spice	How to place
Voltage Source	Voltage (V)	



		Press F2, or Click this icon then search for voltage
Inductor	Inductor (L)	Press F2, or click that icon and search ind
Diode	Diode(D)	Press F2, or click that icon and search diode
MOSFET	MOSFET (M)	Press F2, or click that icon and search nmos
Capacitor	Capacitor (C)	Press F2, or click that icon and search cap
Resistor	Resistor (R)	Press F2, or click that icon and search res
Ground	GND	Press F2, or click that icon and search gnd

Important:

After selecting the component, **click anywhere on the schematic to place it.**

Step 4: Place Two Voltage Sources

You must place **two voltage sources**.

V1: Input Supply

This represents the **5V input supply**.

V2: PWM Signal.

Step 5: Arrange the Components

Arrange the components like the circuit diagram.

The connection order is very important.

Input Section

Connect: V1 positive to MOSFET Drain node

This node will also connect to:

- Inductor L1
- Diode D1

Inductor Connection

Connect: Top of Inductor (L1) to same node as MOSFET drain.

Bottom of Inductor (L1) to Ground

Diode Connection

Diode has two terminals:

- Anode (A)
- Cathode (K)

Connect: Anode to Inductor/MOSFET node

Cathode to Output node

Output Capacitor

Connect: Positive terminal of C1 to Diode cathode

Negative terminal of C1 to Ground

Load Resistor

Connect: R1 must be connected parallel to the capacitor.

Meaning: Top of R1 to Output node

Bottom of R1 to Ground

MOSFET Connections

MOSFET has three pins:

Pin	Connection
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Drain (D)	Input node with inductor
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Source (S)	Ground
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Gate (G)	PWM source
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Connect:

Gate to PWM voltage source (V2)

Step 6: Connect Wires

- Press **F3** to start wiring.
- Click on a pin to drag to connect to another component.

Carefully wire:

- V1 to MOSFET drain node
- MOSFET drain to Inductor

- MOSFET drain to Diode anode
- Diode cathode to Capacitor positive
- Capacitor negative to Ground
- Resistor across capacitor
- MOSFET source to Ground
- PWM source to MOSFET gate

Step 7: Set Component Values

Component Value

V1	DC = 5V
V2 (PWM)	PULSE source
L1	330 μH
D1	1N5819 (Schottky)
MOSFET	Si7336ADP
C1	470 μF
R1	20 Ω

Step 8: Configure PWM Source

Right-click the PWM voltage source.

Select **Advanced** → **PULSE**.

Fill these values:

Parameter Value

Vinitial	0
Von	5
Tdelay	0
Trise	30ns
Tfall	30ns

Parameter Value

Ton 40μs

Tperiod 100μs

Note: This project works on buck boost topology So up to 50% of duty will work for Buck Converter & up to 99% of duty will work for Boost converter.

Step 9: Add Ground Connections

Press **F2** → **GND**

- Connect ground to:
- Negative terminal of **V1**
- Negative terminal of **PWM source**
- Bottom of **inductor**
- MOSFET **source**
- Capacitor negative terminal
- Resistor bottom terminal

Step 10: Setup Simulation

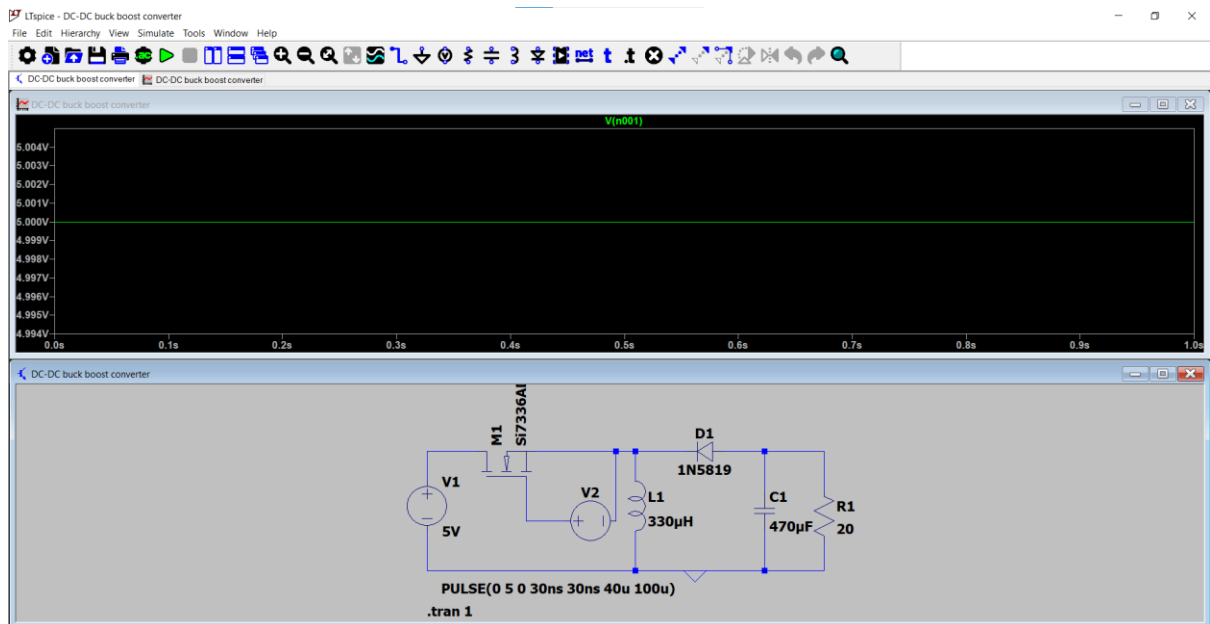
- Press S or go to simulate > Edit simulation Cmd.
- Choose Transient tab.
- Set stop time:
 - 1) Stop time: 1ms
- Click Ok.
- Place the. **tran 1ms** command on the schematic.

Step 11: Save and Run Simulation

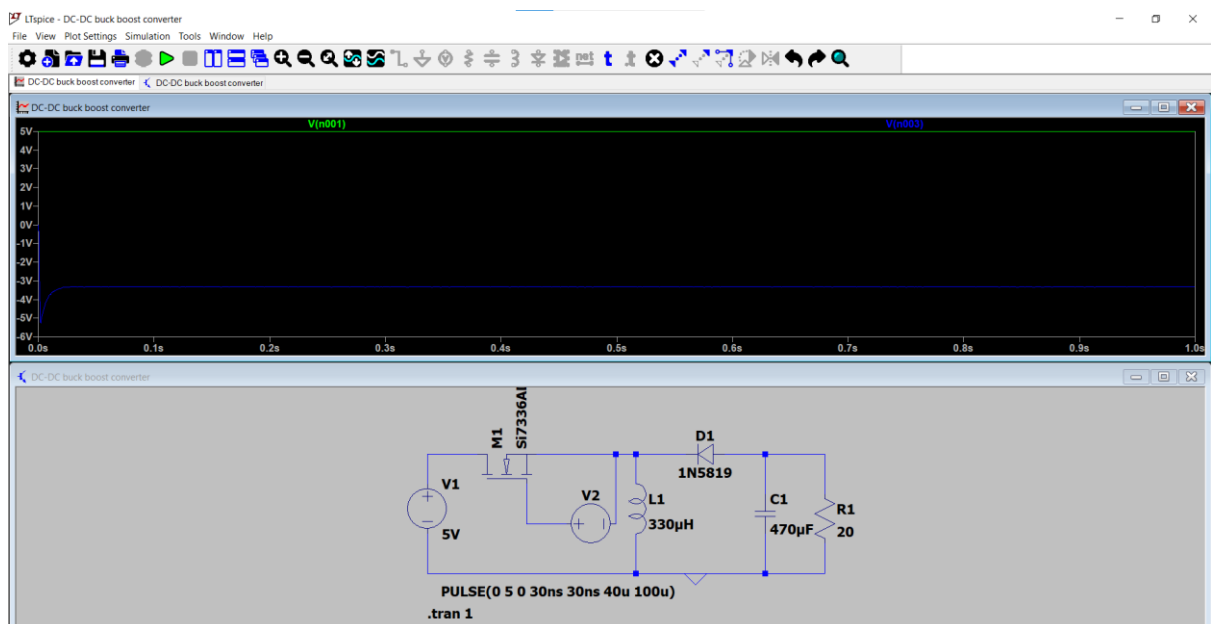
- Click save.
- Then click the Run button.
- The simulation will start.

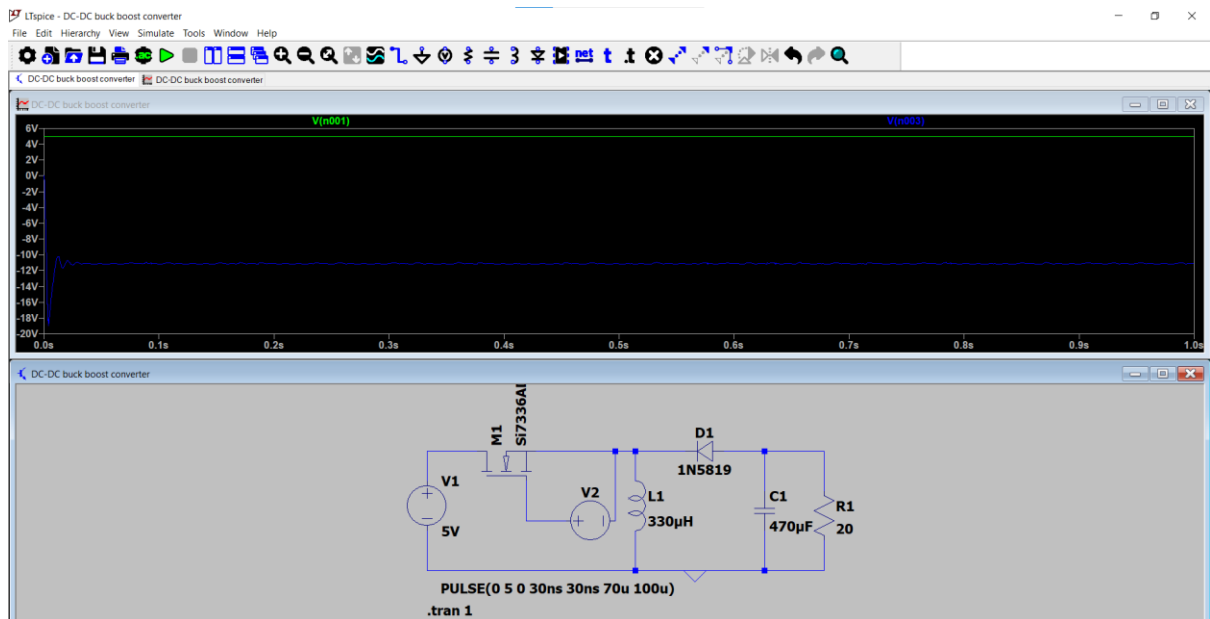
Expected Output:

Input voltage:



Buck Conversion Output:





Tasks:

- 1) Set the input voltage as 12V and design a Buck Boost Converter to get an output of around 24V and 5V.
- 2) Simulate the same Buck Boost Converter circuit with different load resistances (e.g., 60 Ω , 120 Ω , 180 Ω) and observe how the output voltage changes.
- 3) Change the inductor value (e.g., 608 μ H to 300 μ H or 900 μ H) and observe how it affects the ripple in output voltage and current.
- 4) Increase or decrease the switching frequency and note how the output voltage ripple and efficiency are affected.
- 5) Plot and observe the Inductor Current, Capacitor Voltage, and MOSFET Drain-Source Voltage during simulation.